

CS3 Rubric – Case Study Create

DS 4002 – Spring 2026 - Creator: Diya Amat

Due: End of Semester

Submission format:

- One 2-4 page PDF report
- One Python script used to perform the analysis
- All work must use the provided GitHub repository materials

Individual Assignment

Preparatory Assignments – Everything in the course, but especially creating your past 3 presentations.

Why am I doing this? This case study gives you hands-on experience with a real-world data science workflow. You will work with real financial and social media data, apply sentiment analysis techniques, perform statistical testing, and interpret noisy, real-world results.

- Course Learning Objective: Understand and apply core data science concepts including:
 - Exploratory data analysis (EDA)
 - Sentiment analysis
 - Statistical hypothesis testing
 - Interpretation of real-world datasets

What am I going to do? You will complete a structured data science mini-project using the materials provided in the GitHub repository. You will:

- Read the scenario and mission in the Hook Document
- Analyze the provided dataset (Reddit + earnings data)
- Perform sentiment and engagement analysis
- Apply statistical testing (Welch's t-test)
- Interpret your findings using visualizations and written analysis

Tips for success:

- Carefully read the Hook Document to understand your role
- Focus on clarity over complexity
- Use visualizations to support your conclusions
- Start early and explore the dataset thoroughly

How will I know I have succeeded? You will meet expectations when your submission clearly demonstrates correct analysis, interpretation, and communication of results based on the criteria below.

Spec Category	Spec Details
Formatting	• Submit a 2–4 page PDF report • Include clear section headers (EDA, Analysis, Results, Interpretation) • Include at least 2–3 visualizations (plots or charts) • Writing must be clear, concise, and organized
Hook Document	You must: ○ Use the scenario to understand your role as a data analyst ○ Follow the high-level mission described in the Hook Document ○ Refer back to it when making decisions about your analysis The Hook document will link you to the GitHub repo and define the goals
Data Analysis/Sentiment & Engagement Analysis	You must: • Load and explore the dataset provided in the repository • Summarize key variables (sentiment, earnings outcome, engagement) • Include at least one EDA visualization (e.g., distribution of sentiment scores) • Compute and compare average sentiment and engagement between earnings beats and misses • Include at least one comparison visualization and describe observed patterns
Statistical Testing	You must: • Perform a Welch Two-Sample t-test • Clearly state: ○ Null hypothesis ○ Alternative hypothesis • Report test statistic and p-value • Interpret whether results are statistically significant
Final Deliverable	• Your PDF report must include: an overview of the dataset, EDA visualizations, sentiment and engagement comparisons, Welch’s t-test results, and a written interpretation of findings. • Your notebook or script must run without errors and reproduce the analysis shown in the report.

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